SHER KHAN PRESENTS

SherWatch Command Center

A network security monitor for Windows, macOS and Linux that shows every program sending data out of your computer, scores it against threat intelligence and known exploits, explains what it sees in plain language, and lets you act on it in one click.

Windows · macOS · Linux | Rust + Tauri | Driverless per-process capture | 28 detection rules | Threat feeds auto-updated | 3D globe | Built-in AI analyst | Runs 100 % locally

Open source · github.com/gokukhan/SherWatch · 2026

WHY NOW

The age of AI: attacks at prompt speed, defence at prompt speed

Attacks at prompt speed

Generative AI has turned writing malware, infostealers and phishing kits from weeks of work into minutes. Attacks are cheaper, faster and more numerous than before, and they reach ordinary computers, not only companies with a security team. The first visible sign is almost always the same: a program quietly sending data out. Most people have no way to see that happening on their own machine.

Defence at prompt speed

SherWatch was designed and directed by Sher Khan and written with Claude Code, Anthropic's AI coding agent, as a pair-programming partner. The tools that make attacks cheap make defence cheap too: a per-process monitor with 28 rules, a live globe and three desktop ports, by one person and an AI pair programmer. It is open source under Apache 2.0, so every rule it applies can be read and checked rather than taken on trust.

WHAT THAT MEANS FOR YOU

- **See** every program that sends data, where it goes, how much, and whether a threat feed or an exploit catalogue knows the destination or the program.
- **Understand** it in plain language: each alert says why it matters and what to do.
- **Act** in one click: terminate, block, trust, and undo any of it from the audit log.
- **Trust it**: it runs entirely on your machine, no account, no cloud, and the source is public.

The analyst assistant inside the dashboard is a separate feature: it explains what the monitor sees, with a model you choose (free web model, local Ollama, or your own key).

WHAT IT DOES

Everything that leaves your PC, who sent it, and whether it should have

See every connection, per program

Live table of all TCP and UDP flows with the owning process, signature, destination, hostname, country, bytes and age. Captured from the operating system's own accounting (ETW on Windows, nettop on macOS, ss on Linux), not a packet sniffer.

Catch what should not be happening

28 rules: threat-feed matches, beaconing, exfiltration, DNS tunnelling, algorithm-generated domains, living-off-the-land tools, masquerading, unsigned software, suspicious locations, new listeners.

Know when a file leaves

Any program not on your approved list that uploads more than a few MB, or uploads to a file-sharing service, raises a critical alert within seconds.

Threat data that keeps itself fresh

Feodo, Spamhaus, IPsum, Tor exits, OpenPhish, Blackbook and CISA's exploited-vulnerability catalogue refresh on a schedule. New sources arrive through a manifest; the app tells you when it needs a code update.

Vulnerable software on this PC

Installed programs and services are matched against CISA KEV and, optionally, published CVEs for the exact version, with a judgement on whether the fix is likely installed.

Act, log, revert

Terminate, block an address in the system firewall (Windows Defender Firewall, a pf table on macOS, an nftables set on Linux), trust a file by hash. Every action is written to an audit log and can be reverted from the Settings page.

ALSO

- **Built-in AI analyst**: ask "what is this process doing?" and get an answer with the facts already attached, from a free keyless model, a local Ollama model or Claude.
- **Interactive 3D globe** of live destinations with per-country colouring and clickable destination cards.
- **Desktop app and browser version** built from the same code for Windows, macOS and Linux, with installers and a tray icon.
- **Local only**: no account, no cloud, the dashboard answers only your own machine.

OVERVIEW

The command center view

SherWatch
COMMAND CENTER

- Overview
- Globe
- Connections
- Processes
- Alerts 210
- DNS
- Inventory
- Assistant
- History
- Settings

Live feed
Packet capture
Administrator

Overview 22:54:42

5 critical · 28 high · Search process, host, IP...

Upload	Download	Connections	Processes online	Open alerts	Session total
2.6 KB/s	11.4 KB/s	282 103 to the internet	72 73 listening ports	210 13 critical · 36 high · 173 medium · 301 info	614 MB ↑ 416 MB · ↓ 198 MB

Live destinations

53 destinations · 52 active



Latest alerts

all

git-remote-https.exe is sending data out of this PC: 20.4 MB in a minute — CRITICAL — 22:46:21
git-remote-https.exe (45996) · R27-file-upload

Unsigned program cbservice.exe is sending data to the internet — MEDIUM — 22:45:57
cbservice.exe (15856) · R04-unsigned · ×9

Unsigned program cbservice.exe is sending data to the internet — MEDIUM — 22:35:57
cbservice.exe (15856) · R04-unsigned · ×10

New application on the network: sdiagnhost.exe — INFO — 22:33:31
sdiagnhost.exe (35480) · R13-new-app

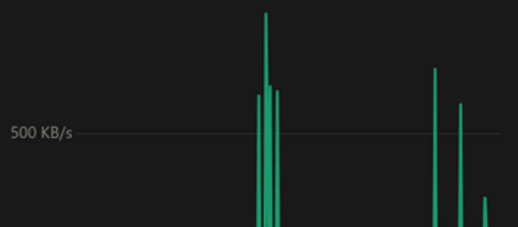
Top talkers

bytes this session



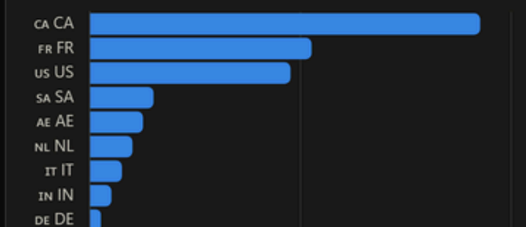
Throughput, last 5 minutes

Upload · Download



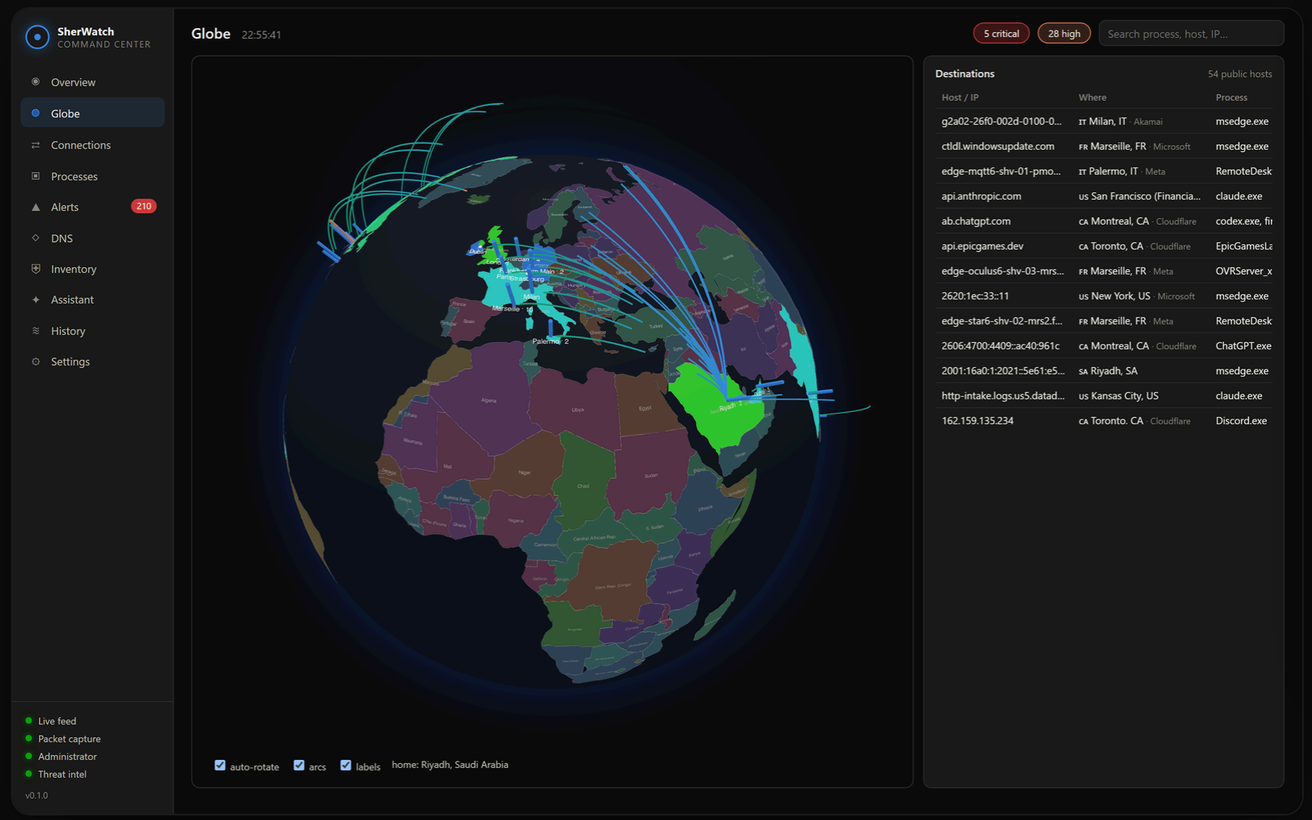
Destinations by country

active connections



GLOBE

Where your data goes, live

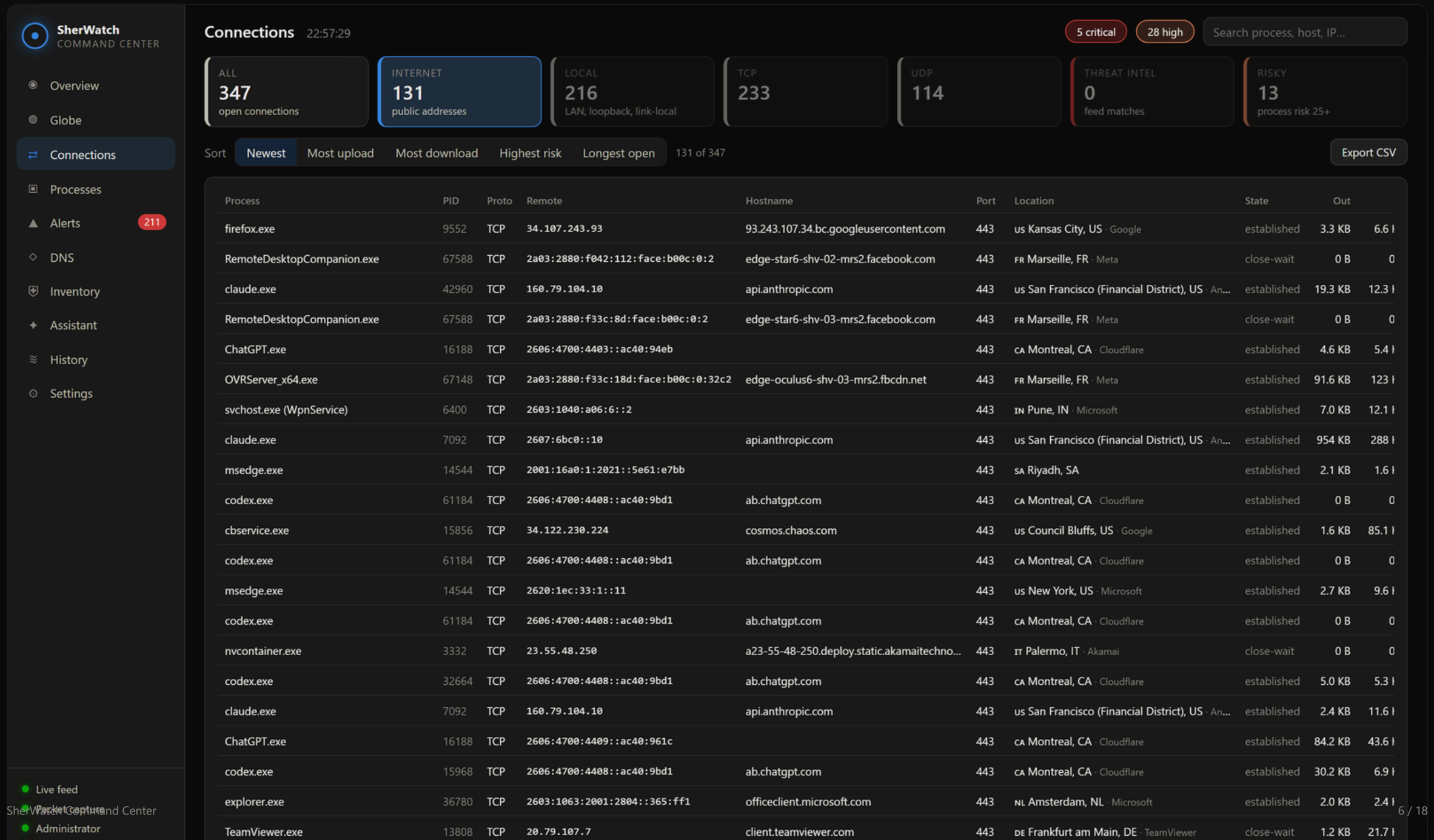




- Arcs from your location to every active destination, coloured by risk; points sized by traffic; one label per city.
- Countries you talk to light up; countries hosting a threat-feed match turn red.
- Hover pauses the rotation. **Click** a point, city or country for the destination card: owner, intel match, bytes and every process sending there, each linking to its drawer, the assistant or the connection list.
- Your home location comes from the config or is detected once from your public IP.

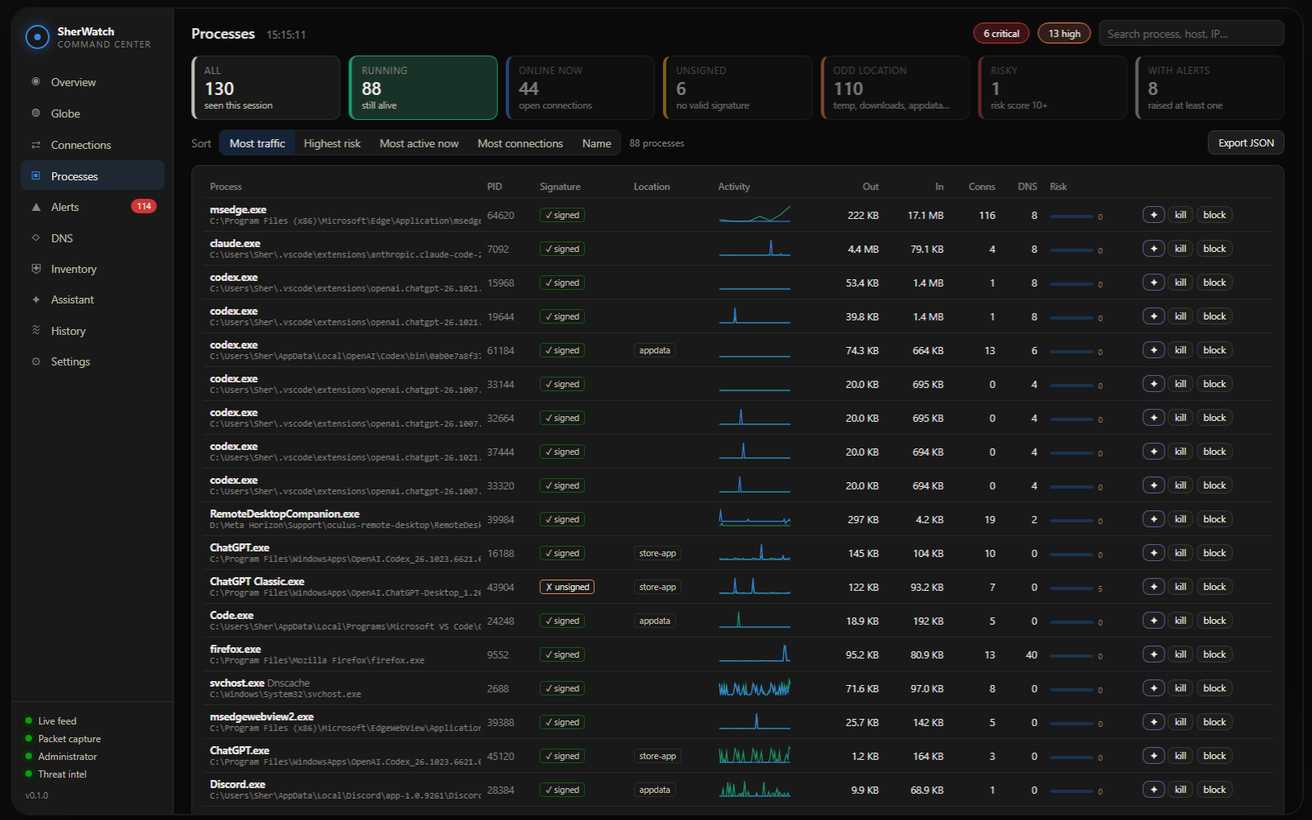
CONNECTIONS

Every live flow with its owner

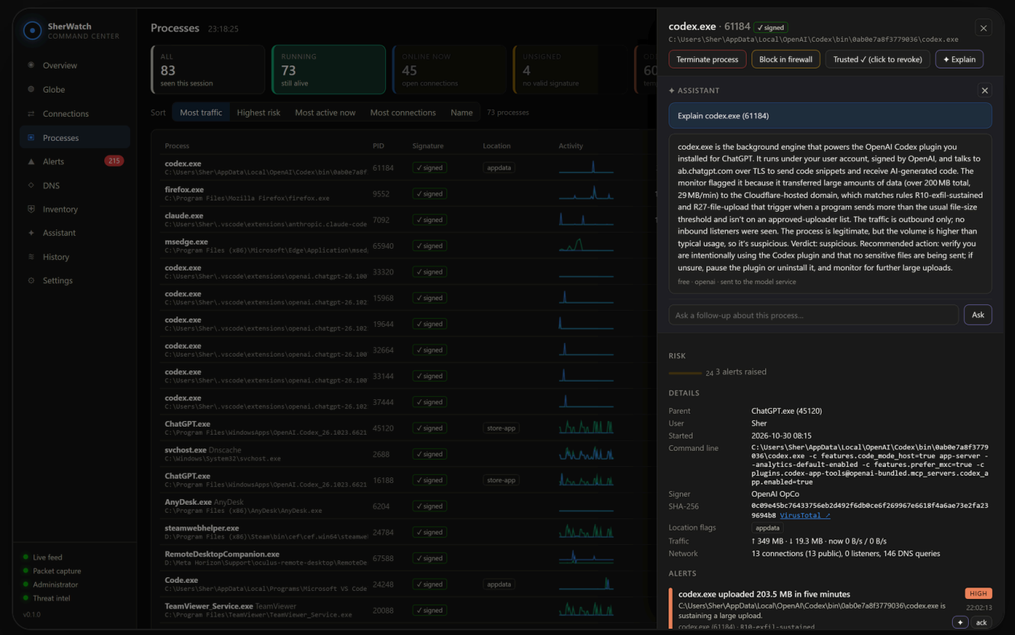


PROCESSES

Who is talking, and the full story of each



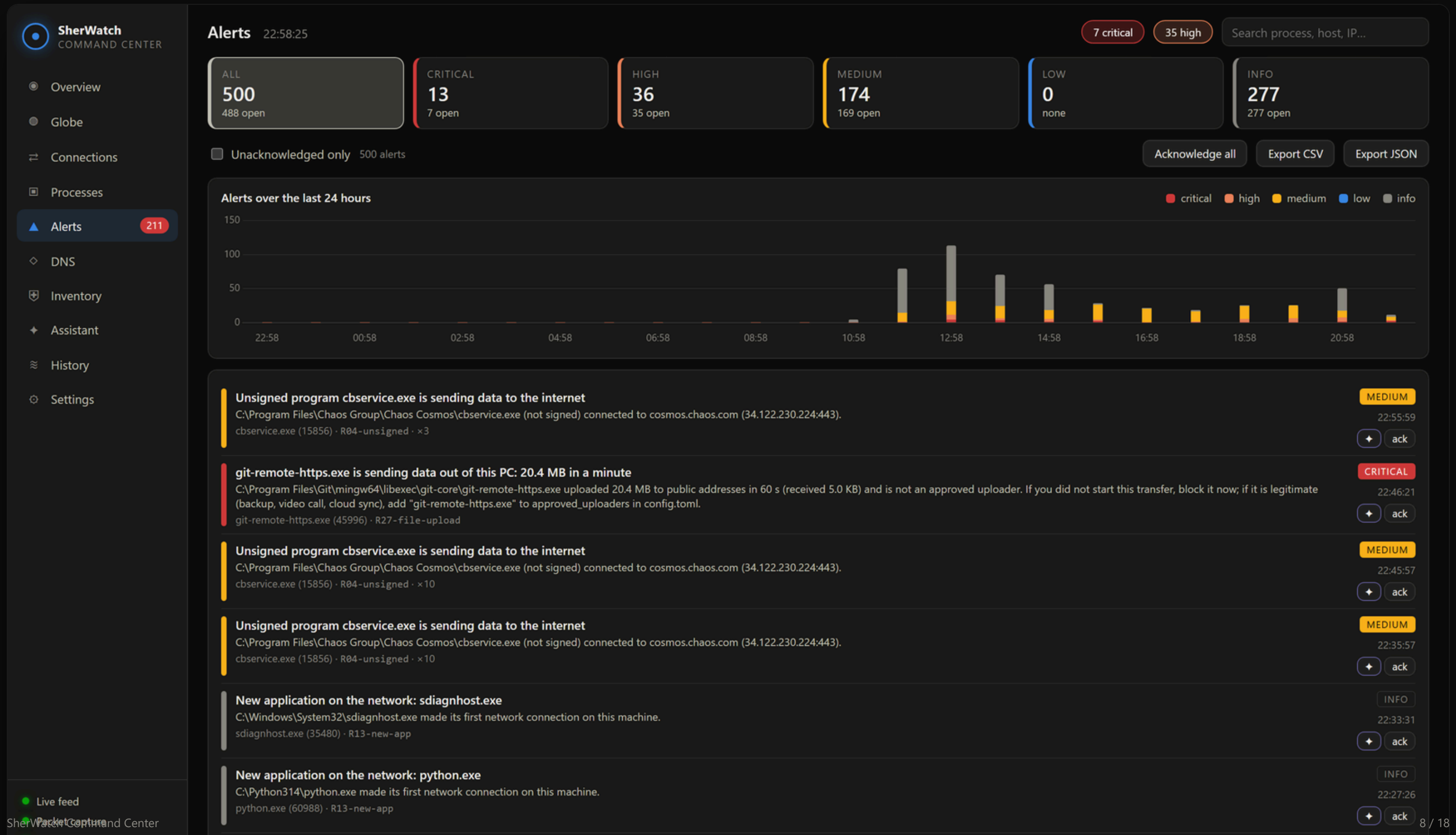
- Cards: running, online now, unsigned, odd location (Temp, Downloads, AppData…), risky, with alerts.
- Signature status with publisher, location flags, 60-second activity sparkline, bytes, connections, DNS queries and a decaying risk score.
- Buttons per row: ✦ explain, kill, block.



- **Process drawer**: parent, user, start time, command line, signer, SHA-256 with VirusTotal link, hosted services, traffic, listening sockets, alerts, connections, DNS and repeated contacts.
- **Terminate**, **Block in firewall** (shows the live state and turns into "allow"), **Trust** by file hash, and ✦ **Explain** with the answer and follow-up box inside the drawer.

ALERTS

Findings, ranked and explained



DETECTION

28 rules, from reputation to behaviour

RULE	SEV	LOOKS FOR	RULE	SEV	LOOKS FOR
R01 / R02	CRITICAL	IP or domain listed in a threat feed	R15	HIGH	Hosts-file hijack or blocked security vendor
R04	MEDIUM	Unsigned or invalid-signature program on the internet	R16	LOW	Connection to a watch-list country
R05	HIGH	Network program in Temp, Downloads, Desktop, Recycle Bin…	R17	MEDIUM	Fan-out: 100+ hosts in a minute (scan, spray, P2P)
R06	HIGH	Living-off-the-land binary (certutil, mshta, rundll32…) online	R18	LOW	Session open for over six hours on an odd port
R07	HIGH	Suspicious destination port: SMB, RDP, IRC, Tor, miners, SOCKS…	R19	HIGH	Network child of Office, Acrobat or a script host
R08	MEDIUM	Program bypassing the system resolver (own DNS)	R20	MEDIUM	Unsigned program using an IP no DNS query explains
R09	HIGH	Beaconing: regular timer-like check-ins	R21	CRITICAL	VirusTotal detections for an unsigned file
R10	HIGH	Large upload volume per minute / five minutes	R22	HIGH	Unsigned program accepting connections from other machines
R11	HIGH	DNS tunnelling patterns	R23	CRITICAL	Process named like svchost, lsass, launchd, systemd… outside the system folders
R12	MEDIUM	Algorithm-generated domain names	R24 / R25	HIGH	Exploited software likely unpatched; OS updates stale
R13	INFO	New application on the network (first time ever)	R26	HIGH	Published CVEs for the exact installed version
R14	HIGH	New listening port (info for signed software)	R27 / R28	CRITICAL	File leaving the PC; upload to a file-sharing service

Severities feed a per-process risk score that decays over time. Every threshold is editable on the Settings page; nothing is hard-coded. Trusting a file by hash silences the signature and location rules for that exact file only. The rules are the same on every platform; the five that read DNS queries (R02, R08, R11, R12, R20) need the DNS capture, which exists on Windows today.

FILES LEAVING THE PC

A critical alert the moment something uploads

R27 · File upload by a non-approved program

Any program not on the approved-uploader list (OneDrive by default) that sends more than the threshold (5 MB in a minute, editable) to public addresses raises a **critical** alert naming the program, the amount and what to do. Backup clients, video calls and AI coding tools will trip it until you approve them deliberately.

R28 · Upload to a file-sharing service

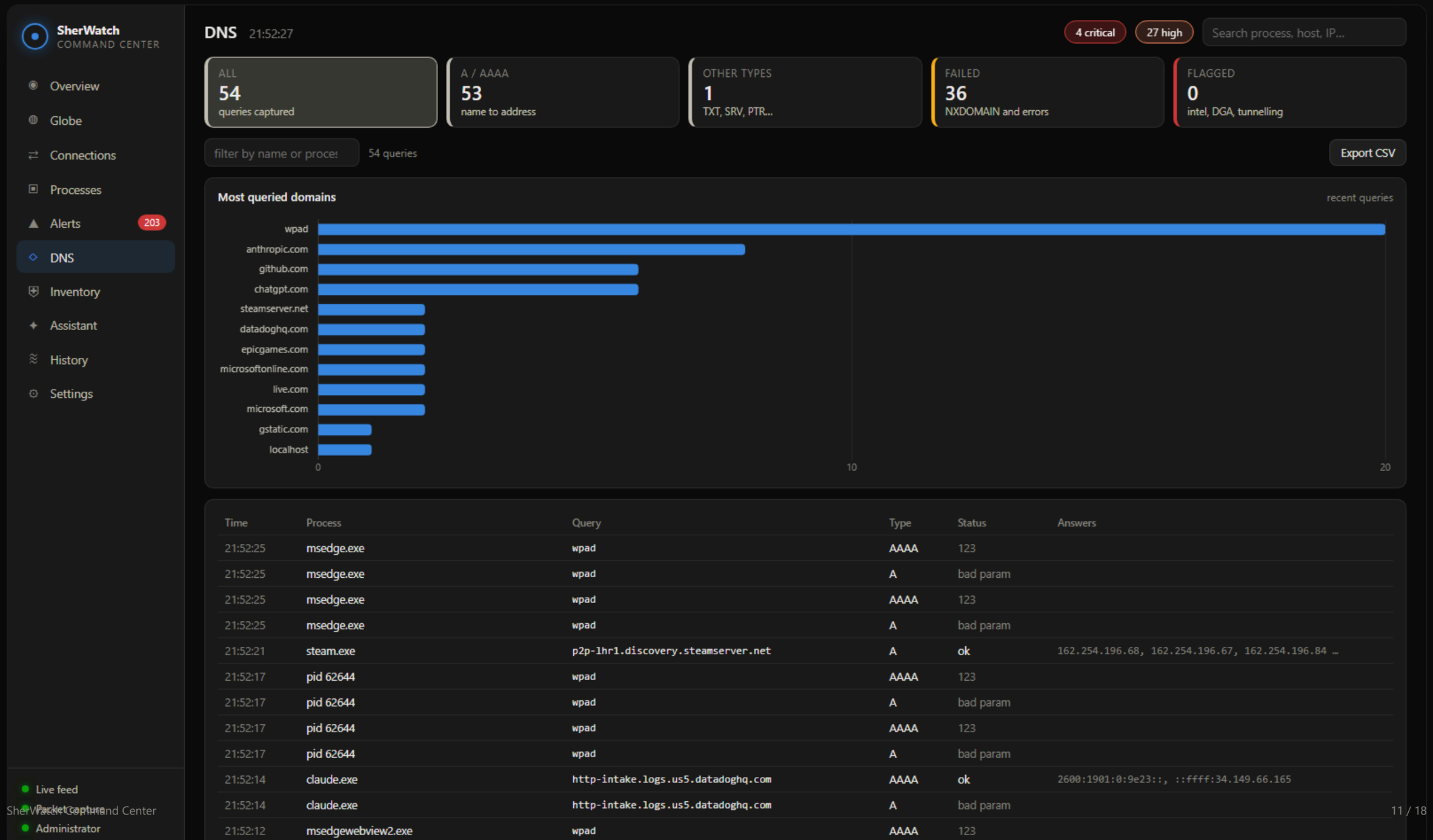
51 built-in services (WeTransfer, Mega, Dropbox, GoFile, transfer.sh, Pastebin, the Telegram API, Discord, S3…) matched by hostname and by address, re-resolved on a schedule. A real upload is critical; merely browsing the site stays silent. The list and threshold are editable.

HONEST LIMITS

- The monitor sees bytes, never file names or content; everything is inferred from behaviour. Traffic is encrypted and is never inspected.
- Loopback traffic is excluded from all volume counts, so the dashboard itself and local dev servers cannot trigger these rules.
- Files leaving by USB stick, Bluetooth, network share, printer or remote-desktop clipboard are not network traffic and are not seen.
- Byte counts need administrator rights; without them R28 drops to a medium "connected to a file-sharing service" notice.

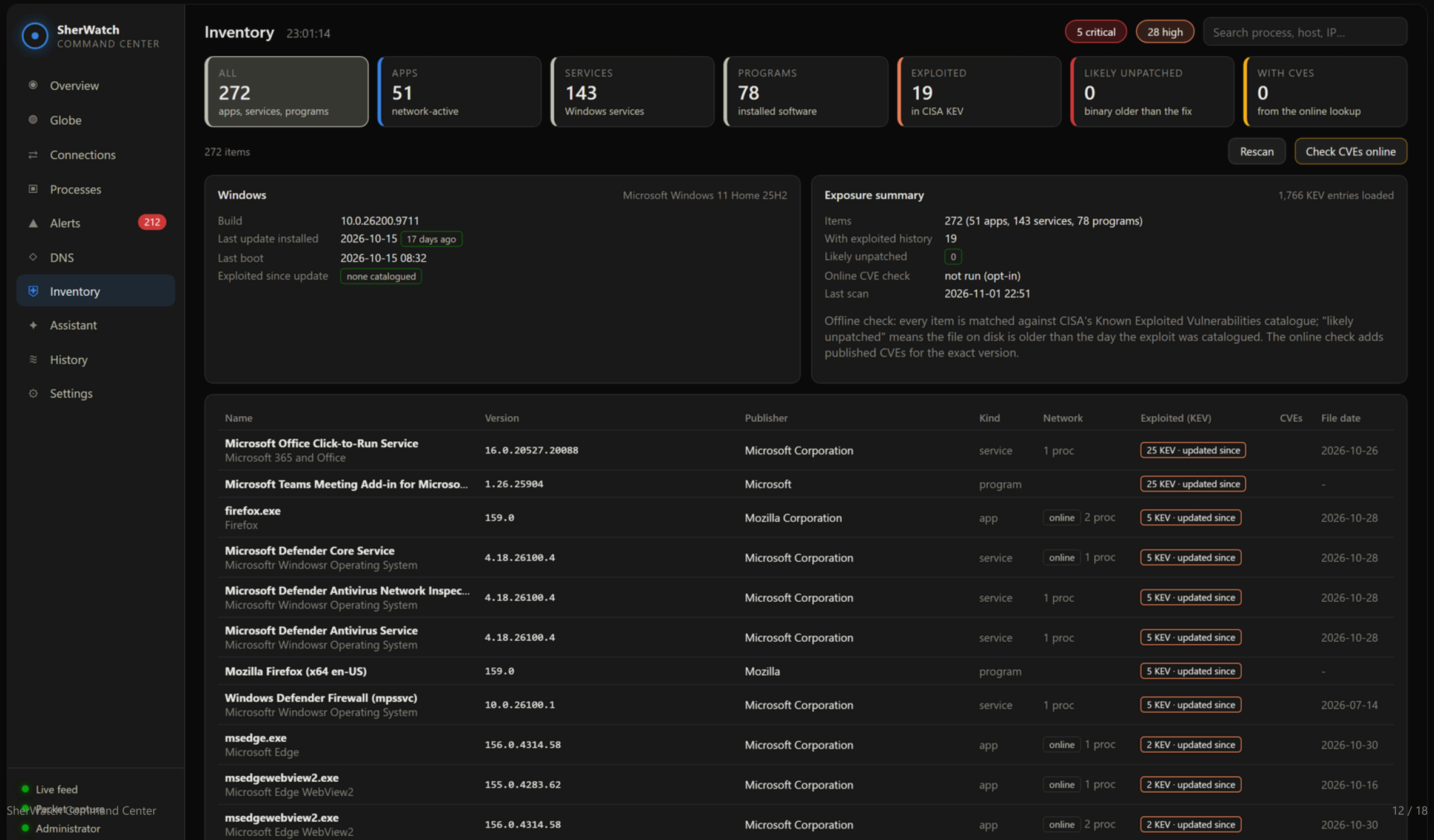
DNS

Every name lookup, per process



INVENTORY

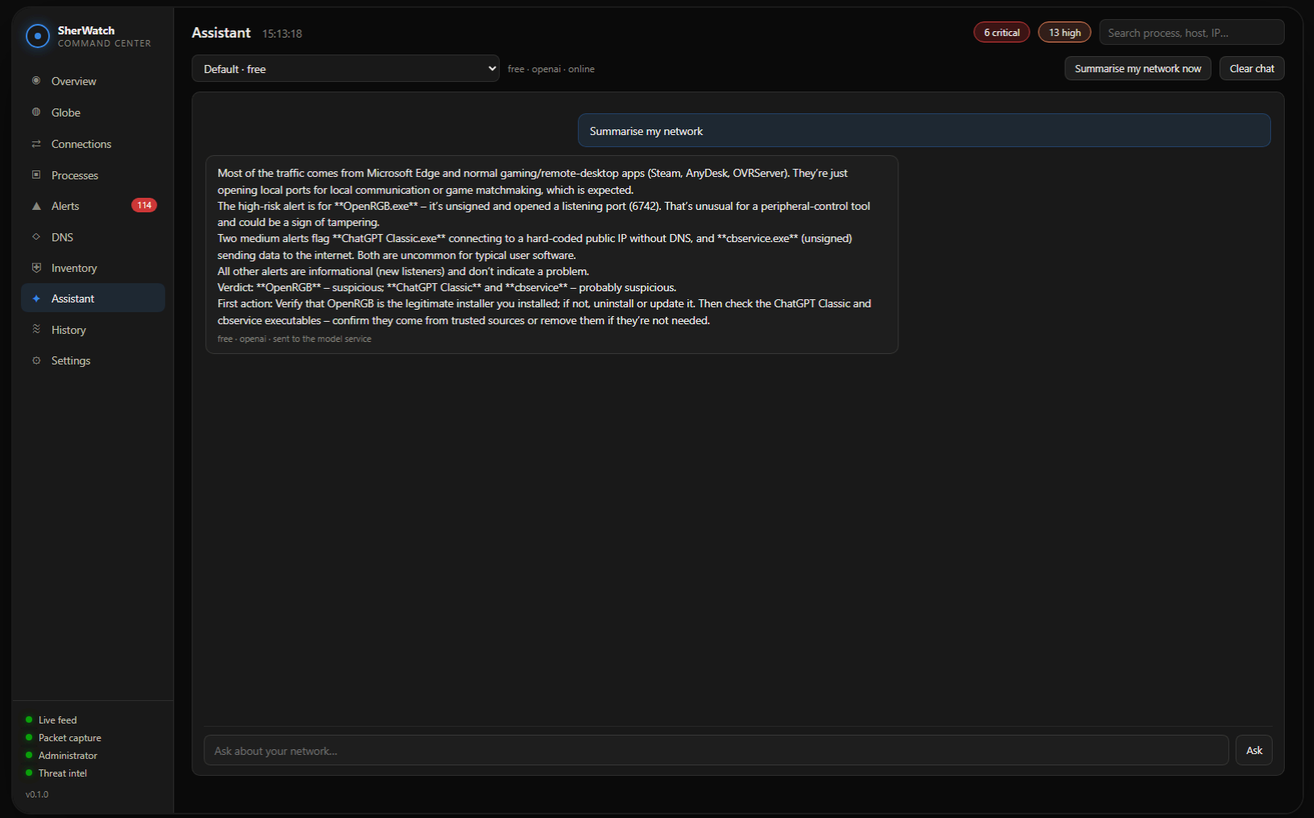
Installed software, services and their known exploits



ASSISTANT

An analyst that already has the facts

- Ask in plain language: "what is svchost 6400 doing?", "is cbservice.exe safe?", "which alerts matter today?", or "summarise my network".
- Every ✦ button attaches the context automatically: path, signature, publisher, parent, hosted services, connections with hostnames and owners, DNS, alerts with rule meanings, vulnerability matches. Answers end with a verdict and a recommended action.
- Answers appear where you are: inside the process drawer with a follow-up box, or in the Assistant view for general questions.
- Choose the model from a dropdown: a free keyless online model by default, any local Ollama model to keep everything on the PC, or Claude and OpenAI-compatible endpoints with your own key.



HISTORY

What happened while you were away

SherWatch
COMMAND CENTER

- Overview
- Globe
- Connections
- Processes
- Alerts 112
- DNS
- Inventory
- Assistant
- History
- Settings

Live feed
Packet capture
Administrator

History 15:02:04

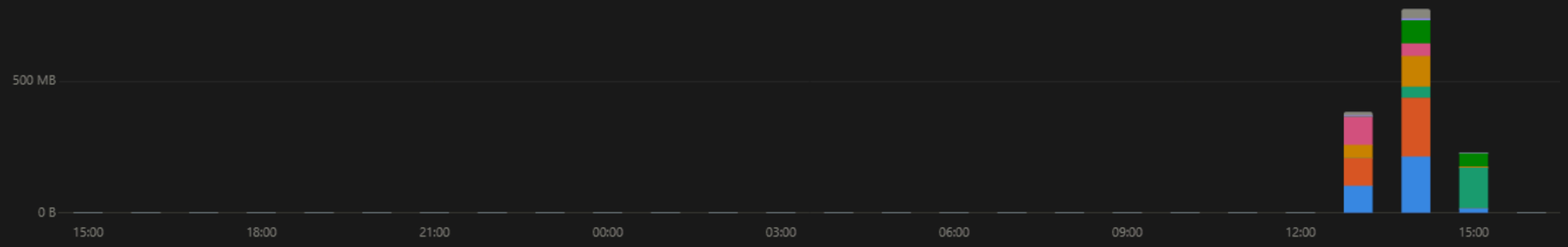
8 critical | 12 high | Search process, host, IP...

1 h | 6 h | 24 h | 7 d | 30 d

stored locally in data/sherwatch.db

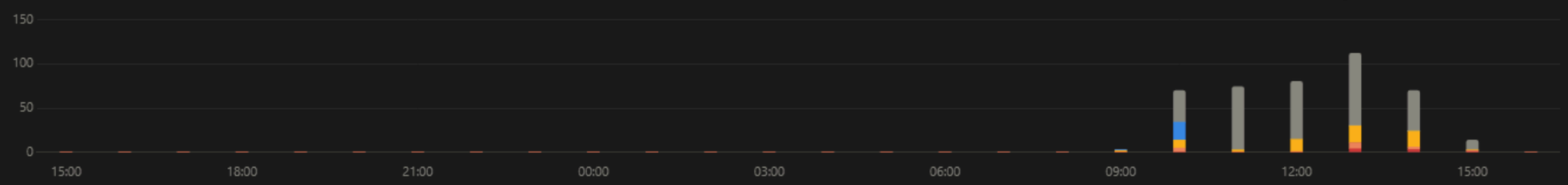
Traffic by application

msedgewebview2.exe | netmon-desktop.exe | msedge.exe | firefox.exe | codex.exe | sherwatch-desktop.exe | claude.exe | Other



Alerts

critical | high | medium | low | info

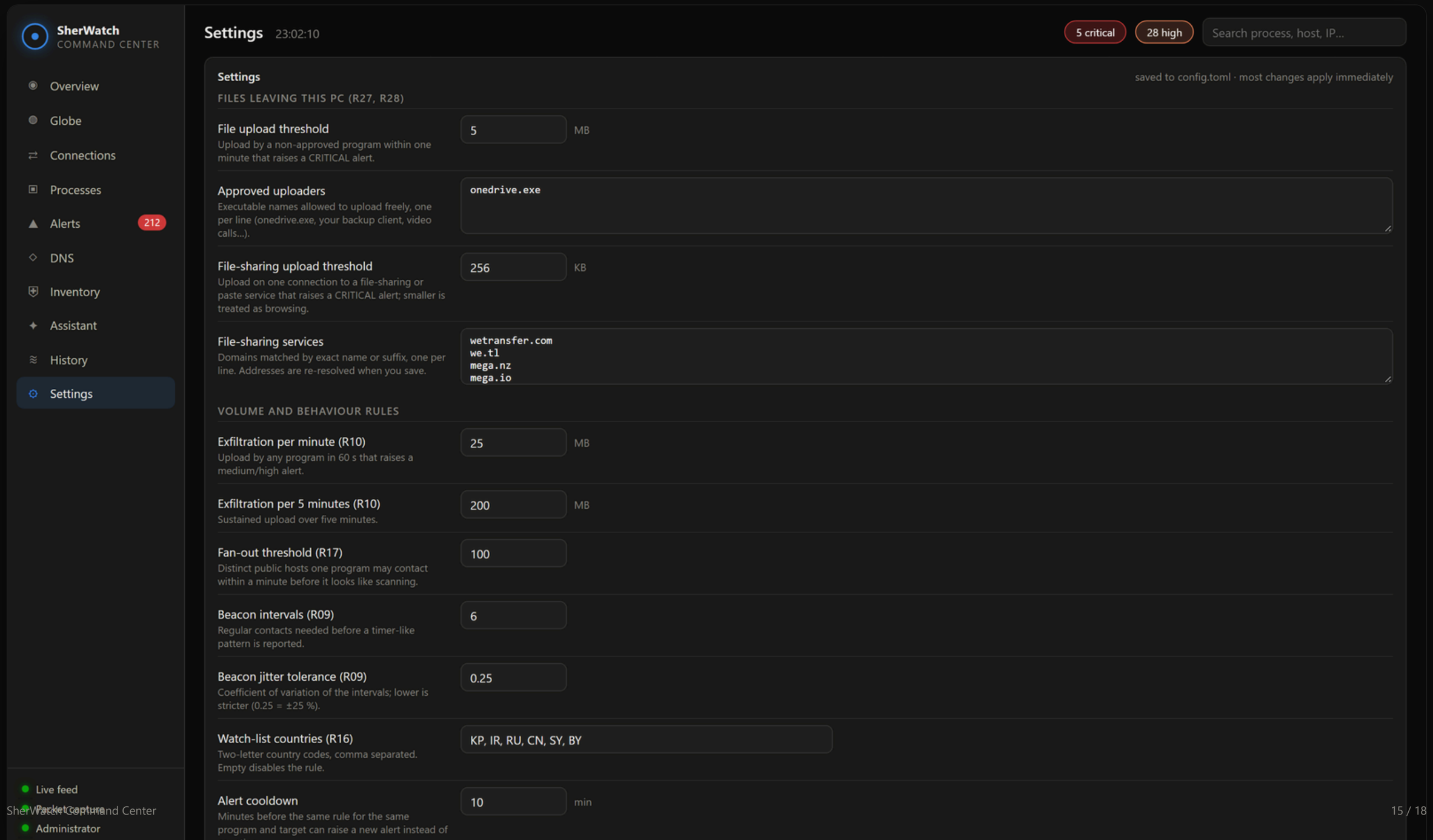


Closed connections 500 shown

Started	Duration	Process	Remote	Hostname	Country	Out	In	Flags
2026-11-01 15:01	1s	msedge.exe	2620:1ec:33::11:443		US US	5.4 KB	58.9 KB	
2026-11-01 15:01	0s	msedge.exe	2603:1047:1:118::80:443		MY MY	5.4 KB	9.3 KB	
2026-11-01 15:01	0s	msedge.exe	2620:1ec:4a::46:443		US US	4.5 KB	12.2 KB	
2026-11-01 15:01	1s	msedge.exe	2a02:26f0:2b80:102::216:fa64:443	g2a02-26f0-2b80-0102-0000-0000-0216-fa64.deploy.st...	FR FR	2.4 KB	22.0 KB	
2026-11-01 15:01	0s	msedge.exe	2a00:1450:4006:81c::2001:443	ncmrsa-aj-in-x01.1e100.net	FR FR	2.4 KB	172 KB	
2026-11-01 15:01	0s	sherwatch-desktop.exe	127.0.0.1:65301			2.2 KB	549 B	
2026-11-01 15:01	0s	msedge.exe	127.0.0.1:8787			544 B	23.0 KB	
2026-11-01 15:01	1s	msedge.exe	2603:1061:10:1::13:443		CA CA	24.0 KB	11.8 KB	
2026-11-01 15:01	0s	msedge.exe	13.107.6.158:443	bingforbusiness.com	US US	4.9 KB	9.7 KB	

SETTINGS AND CONTROL

Tune it, act on it, undo it



UNDER THE HOOD

Built for speed and a small attack surface

Collectors

Per-process bytes from the OS's own accounting: the kernel network trace (ETW) on Windows, nettop on macOS, ss on Linux; socket tables polled every second; process inspector with signature checks (Authenticode, codesign, package ownership) and hashing; reverse DNS; DNS capture on Windows; optional VirusTotal.

Analysis

Rules run every two seconds over a snapshot: feeds, GeoIP, timing windows for beacons and fan-out, DNS windows, inventory matching every 30 minutes.

Storage and intel

SQLite with a batching writer; feeds and the KEV catalogue cached on disk and refreshed on a schedule; manifest-driven feed additions.

Front ends

One Rust library, two thin binaries: a browser version and a Tauri desktop app with tray icon and single instance, packaged as an NSIS installer, a .dmg and .deb/.AppImage. Same dashboard, no drift.

SECURITY DESIGN

- Written in Rust; the UI is plain HTML with no framework and renders only escaped data. All OS-specific code lives in one folder (src/platform); the rules never touch an OS API.
- The API answers only the local dashboard: Host and Origin are checked on every request, which defeats DNS rebinding and cross-site requests from other tabs.
- The desktop window loads the local server like a browser tab and is granted no Tauri commands, so a rendering bug cannot reach the operating system.
- External tools run without consoles; the monitor exempts only its own process from rules that its own behaviour would trip.
- Nothing leaves the PC except feed downloads, the one-time public-IP lookup and the assistant requests you choose to send.

PLATFORMS

Windows · macOS · Linux

	WINDOWS	MACOS	LINUX
Connections per process	OS TCP/UDP table	libproc	/proc
Byte counts	ETW kernel-network trace, driverless	polling nettop	polling ss (iproute2), TCP only
DNS capture	ETW DNS-Client	not yet	not yet
Elevation	UAC prompt, relaunches itself	sudo (run.sh) for all-process byte counts; desktop app unprivileged, only firewall actions prompt	sudo (run.sh) for other users' sockets and bytes; desktop app unprivileged, only firewall actions prompt
Code signature	Authenticode, incl. catalog-signed files	codesign (Apple, Developer ID)	package ownership (dpkg/rpm), not a cryptographic check
Block	IP and program, Windows Defender Firewall	IP, pf table (anchor com.sherkhan.sherwatch); no per-program rules	IP, nftables set (iptables fallback); no per-program rules
Inventory and exploit exposure	services, installed programs, hotfix age; Windows + apps in CISA KEV	launchctl, system_profiler, softwareupdate history; macOS + apps	systemd, dpkg/rpm/flatpak, package-manager age; kernel + apps
Desktop package	NSIS installer (WebView2)	.dmg, unsigned: right-click, Open	.deb / .AppImage (needs libwebkit2gtk-4.1, libayatana-appindicator3)

DOWNLOADS · GITHUB.COM/GOKUKHAN/SHERWATCH/RELEASES

WINDOWS `SherWatch_0.1.0_x64-setup.exe`

MACOS `SherWatch_0.1.0_aarch64.dmg` `SherWatch_0.1.0_x64.dmg`

LINUX `SherWatch_0.1.0_amd64.deb` `SherWatch_0.1.0_amd64.AppImage`

One code base: the rules, the dashboard, the database and the assistant are shared; only the folder `src/platform` differs per OS. GitHub Actions checks every push on all three systems and builds the five packages for every tagged release.

GETTING STARTED

Two ways to run it

Desktop app

Install the package for your system: the `-setup.exe` on Windows, the `.dmg` on macOS (first launch: right-click, Open), the `.deb` or `.AppImage` on Linux. Start SherWatch like any app; on Windows accept the administrator prompt. The window opens on the Overview; closing it keeps monitoring in the tray, Quit stops it.

Browser version

Run `run.cmd` (Windows) or `./run.sh` (macOS and Linux, asks for sudo) in the project folder. The server starts and the dashboard opens at `http://127.0.0.1:8787/` in your browser.

FIRST MINUTES

- The GeoIP database (about 120 MB) downloads once; threat feeds refresh in the background. Everything works meanwhile.
- Administrator rights (sudo on macOS and Linux) unlock byte counts and the file-exfiltration rules, and on Windows the DNS capture; without them the connection view still works.
- Expect a burst of info alerts on the first run as every program you use is seen for the first time; after that only new things are reported.
- When your own backup, call or AI tools trip the file-upload rule, approve them in Settings and they stay quiet.

DOCUMENTATION

User guide, detection rules, architecture, building, updating and third-party notices live in the `docs/` folder of the public repository `github.com/gokukhan/SherWatch`. SherWatch is open source under the Apache License 2.0: clone it, build it, read every rule; free for personal and commercial use, keep the credit.

Available for Windows, macOS and Linux.